

Fear of the dark: decline in plant diversity and invasion of alien species due to increased tree canopy density and eutrophication in lowland woodlands

Martin Vojík  · Karel Boublík

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Abstract The abandonment of traditional forest management and the conversion of lowland woodlands from coppices to high forests may lead to a reduction in plant diversity. We studied long-term changes in semi-natural lowland woodland vegetation in the Czech Republic (Central Europe) by resampling 29 vegetation plots (relevés) first sampled in the 1950s. The results indicated a shift to shady plant communities (i.e. loss of heliophilous species and expansion of shade-tolerant species), caused by an increase in the tree and shrub layer canopy due to the expansion of deciduous tree species. At the same time, species richness (alpha diversity) had declined significantly, though we noted no signs of homogenisation

in the vegetation cover. Species typical of nutrient-rich habitats and alien and/or invasive species had expanded. We confirmed the shift in vegetation composition, which was probably caused by changes in lowland woodland forest management (decline in coppicing), eutrophication (deposition of atmospheric nitrogen, agricultural runoff) and disturbance effects of game, especially wild boar (*Sus scrofa*).

Keywords Biodiversity · Deciduous forests · Plant invasion · Semi-permanent plots · Vegetation changes

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M. Vojík (✉)
Department of Invasion Ecology, Institute of Botany, The
Czech Academy of Sciences, 252 43 Průhonice,
Czech Republic
e-mail: martin.vojik@ibot.cas.cz

M. Vojík · K. Boublík
Faculty of Environmental Sciences, Czech University of
Life Sciences Prague, Kamýcká 129,
165 21 Praha 6 – Suchbátka, Czech Republic
e-mail: boublik@fzp.czu.cz

Introduction

Lowland woodlands (up to 400 m a.s.l.) have been extensively used as a source of wood and other raw materials since the Neolithic period (Hédl et al. 2011). As such, woodlands have been influenced by ‘traditional’ forms of management, such as coppicing for firewood (Szabó 2010), for thousands of years. The general abandonment of traditional coppice management in the mid-20th century has led to a reduction in both animals and vascular plant diversity (Szabó 2010; Hédl et al. 2010; Sebek et al. 2015). The collection of leaves for farm animal litter has been shown to promote plant species richness, especially for annual plants (Vild et al. 2015; Douda et al. 2017). Conversion of lowland woodland from coppice to high forest

has increased tree canopy density and shading, however, resulting in a decrease in heliophilous species in the herb layer (Baeten et al. 2009). Central European lowland woodland species composition has also been affected by wild herbivores (Verheyen et al. 2012), which prevent tree regeneration and reduce tree species diversity (Petřík et al. 2009). Activity of game in forest plant communities has also been shown to support expansion of nitrophilous plant species (Hédl et al. 2010). Note, however, that herbivores may also have a positive effect on species diversity in oak-hornbeam forests (*Quercus petraea*, *Carpinus betulus*), as grazing can reduce shrub layer density and provide openings for heliophilous species (Petřík et al. 2009; Čermák 2011; Alfonsi et al. 2016; Vild et al. 2017).

Studies on vegetation changes in lowland woodlands following abandonment of past management practices have taken place in both Europe and North America (Oakes et al. 2014; Becker et al. 2017; Heinrichs and Schmidt 2017), with the spread of shade-tolerant, mesophilous and nitrophilous species most often documented (e.g. Lameire et al. 2000; Verheyen et al. 2012; Kopecký et al. 2013; Reinecke et al. 2014; Becker et al. 2017; Reczyńska and Świerkosz 2017). Hédl et al. (2010) and Kopecký et al. (2013), for example, observed a decline in overall species richness (alpha diversity), while Brewer (1980) documented the loss of forest herb generalists. Rogers (2006) detected spatial homogenisation and the expansion of alien species within lowland woodland stands. On the other hand, Reczyńska and Świerkosz (2017) noted that changes in species composition were not necessarily dependent on type of forest management, with von Oheimb and Brunet (2007), for example, noting that unexpected diseases, pests or climatic extremes could also play a decisive role in forest dynamics.

Long-term vegetation monitoring is an important tool for observing and predicting the dynamics of plant invasions in (semi-)natural communities (Petřík et al. 2009). Plant invasions and alien species introductions can have a significant impact on plant community diversity (Pyšek et al. 2004), the negative influences being manifested through a decline in species richness, which reduces the natural value of the stands (Hédl 2004a).

The aim of this study was to determine changes in vegetation composition and plant diversity in lowland forest vegetation stands over the last 60 years using

repeat vegetation surveys. At the same time, changes were assessed in relation to environmental conditions. In agreement with the above-cited literature, we hypothesise decrease in species richness (alpha diversity) especially due to the loss of heliophilous species in the herb layer caused by more closed tree layer (decline of light availability in undergrowth). In connection with the decrease of species richness and changes in forest management practices, we expect structural homogenisation between stands (i.e. significant decrease of beta diversity). Moreover, we propose that wild game activity may lead to the spread of nitrophilous and alien or invasive species (from nearby settlements or forest roads), which can significantly jeopardise the conservation of biodiversity. Wild game activity can also facilitate the spread of ruderal species in forest stands.

Results from our long-term (60 years) study can contribute to understanding of current biodiversity loss and change in species composition in lowland woodlands and to general vegetation changes (eutrophication, spread of nitrophilous, alien and generalist species, loss of specialists).

Materials and methods

Study area

This study took place in the Klíneč forest district in Central Bohemia, Czech Republic (Fig. 1). The study

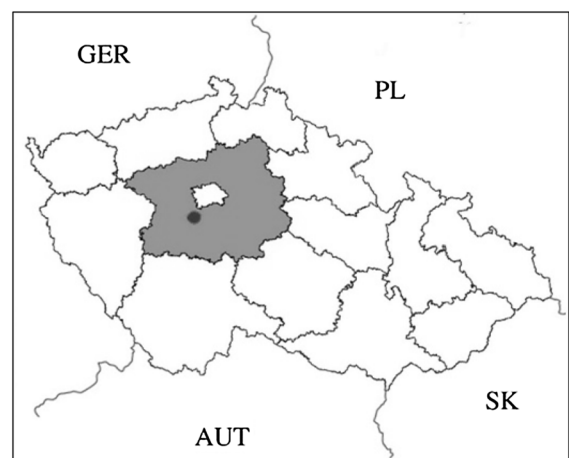


Fig. 1 Location of the study area, southwest of Prague in the Czech Republic

area covers a total area of ca. 1000 hectares, with altitude ranging between 210 and 490 m a.s.l. (Samek 1957). The local climate is warm and dry with a mean annual temperature of 7–8 °C and 550–600 mm annual precipitation (Tolasz et al. 2007). The southern part of the study area comprises quartz porphyry, while slate dominates in the northern part. Cambisol soils are predominant, with steep hillsides being covered with leptosols (Culek et al. 2013). Historically, most of the study area has been influenced by coppice management along with unlimited sheep grazing (Samek 1957).

Reference records

Archival reference records (in the form of phytosociological relevés [plots]) were produced by the Czech forest ecologist and botanist V. Samek in 1957. Note, however, that these early studies were somewhat subjective as Samek tried to capture all variability in vegetation over the study area (see Samek 1957). Each relevé was originally defined by local forestry territories (each forestry territory is defined by several hectares) and were depicted in relevant forestry maps of the period. As not all relevés were defined with sufficient accuracy, they have usually been designated as semi-permanent plots (Naaf and Wulf 2011; Reinecke et al. 2014; Savage and Vellend 2015; Kapfer et al. 2016).

Relocation of old relevés

The precise location of old plots was determined using georeferenced archival forestry maps at a scale of 1:10,000, using a digital terrain model (hereinafter DTM) of the area (Farr et al. 2007), with each forestry territory digitised in vector format in ArcGIS (ESRI 2011). Altitude, slope and aspect value was then calculated from the DTM. Potential inaccuracies in altitude, slope and aspect measurements for the old plots were reduced using predefined tolerance values (altitude ± 20 m, slope $\pm 5^\circ$, aspect $\pm 45^\circ$). Twenty-nine 500 m² circular plots met the aforementioned criteria and these were used as the basis for all subsequent resampling.

Resampling method

The 29,500 m² plots were resampled by the first author (MV) during the growing season of 2015 using the methodology of Mráz and Samek (1955), i.e. the same methodology applied in the original research of Samek (1957). Data from a single author (Samek 1957) were used in order to limit errors arising from the use of data by more authors (different professional experience, qualities of taxa determination, etc.). Information on tree layer composition was also used to ensure precise localisation of the plots and to reduce errors from misplacement (Kopecký and Macek 2015). The use of relatively large relevés (i.e. 500 m²), and further surveying in the relevé surroundings, limited the likelihood of pseudo-turnover (Nilsson and Nilsson 1985; Fischer and Stocklin 1997; Giarrizzo et al. 2017). Phanerogam nomenclature follows that of Danihelka et al. (2012). The list of Pyšek et al. (2012) was used for detection of alien species.

Data handling and editing

Data on the ‘new’ relevés were stored in the Turboveg for Windows database program (Hennekens and Schaminée 2001), with each new relevé assigned to the respective old relevé. The dataset was subsequently edited in JUICE 7.0 (Tichý 2002). Related taxa in the dataset were merged into aggregates (agg.). The complete dataset listing old and new relevés is provided as an electronic supplement (see Supporting Information Table S1).

Data analysis

In order to determine overall changes in vegetation, we compared the fidelity of species to the old and new set of relevés, respectively, with fidelity calculated using the phi coefficient (significance level of Fisher’s exact test $p < 0.05$). Overall changes in vegetation for each relevé pair were visualised using a non-metric multidimensional scaling (NMDS) ordination in two dimensions. We performed global NMDS on Bray–Curtis dissimilarity matrices using the JUICE-R function and R v. 2.9.0 (R Development CoreTeam 2010) with the *isoMDS* (MASS package) (Ripley et al. 2002). NMDS diagram was supplemented by passively projected Ellenberg indicator values (EIV).

These diagrams highlight significant ($p < 0.001$) vegetation shifts along basic ecological gradients (i.e. light, moisture, soil reaction, nutrient and continentality).

Alpha diversity (species richness) was expressed as the number of species in each plot and tested using the paired t test. Changes in composition heterogeneity (beta diversity) were assessed using a Bray–Curtis distance matrix on square-root transformed data in a principal coordinate analysis (PCoA). The distance of plots to their habitat group centroid (file median) in the first and second survey was calculated using analysis of multivariate homogeneity of group dispersions ‘betadisper’ in ‘vegan’ R package (PERMDISP; Anderson 2006) and displayed in PCoA diagrams. Since Bray–Curtis dissimilarity is not Euclidean, the distances from axes of PCoA were calculated employing the procedure for the analysis of multivariate homogeneity of Anderson (2006).

Changes in environmental conditions were determined through a comparison of mean EIV values unweighted by species cover (calculated in the JUICE program) for each relevé (Navrátilová et al. 2017), with changes in mean EIV between old and new relevés tested by analysis of variance (ANOVA).

Results

Overall change in vegetation composition

Species of shallow soils (e.g. *Danthonia decumbens*, *Calluna vulgaris*, *Genista tinctoria*, *Veronica officinalis*), acidophytes (*Avenella flexuosa*, *Luzula luzuloides*) and oak-hornbeam forest species (e.g. *Galium sylvaticum*, *Hepatica nobilis*) were frequently recorded in the old dataset, but a clear decrease in frequency was noticed in the new dataset. *Pinus sylvestris* has been replaced by *Carpinus betulus*, *Larix decidua* and *Quercus petraea* in the tree layer, while nitrophilous plants such as *Impatiens parviflora* and *Rubus fruticosus* agg. became more frequent in the herb layer (Table 1).

The NMDS diagram (Fig. 2) shows a decrease in light intensity (i.e. increased shading) to be the main trend in vegetation change over the 60-year study period. In addition, there has been a shift toward higher soil moisture, increase in nutrient content and reduction in acidophilus species presence (species

Table 1 List of species that increased or decreased the most in abundance when comparing old and new relevés

Increased	Decreased
<i>Quercus petraea</i> (E ₃)	<i>Veronica officinalis</i>
<i>Quercus petraea</i> (E ₁)	<i>Luzula luzuloides</i>
<i>Carpinus betulus</i> (E ₃)	<i>Picea abies</i> (E ₃)
<i>Picea abies</i> (E ₁)	<i>Fragaria vesca</i>
<i>Impatiens parviflora</i>	<i>Hieracium murorum</i>
<i>Rubus fruticosus</i> agg.	<i>Pinus sylvestris</i> (E ₁)
<i>Fagus sylvatica</i> (E ₂)	<i>Veronica chamaedrys</i>
<i>Sorbus aucuparia</i> (E ₁)	<i>Anemone nemorosa</i>
<i>Larix decidua</i> (E ₃)	<i>Potentilla erecta</i>
<i>Melampyrum pratense</i>	<i>Moehringia trinervia</i>

showing highest fidelity between old and new relevés are indicated in Table 2). This main trend in species composition has occurred especially within dry and nutrient-poor vegetation. The only exception were sites rich in nutrients and soil moisture already 60 years ago (e.g. scree and floodplain forests). Conditions of these sites have remained largely unchanged as indicated by short trajectories between old and new plots (Fig. 2).

Overall, strictly acidophilus species (*Avenella flexuosa*, *Calluna vulgaris*, *Genista tinctoria*) showed high fidelity to the old relevés, whereas slightly acidophilus taxa (*Melampyrum nemorosum*) were more typical for new relevés (Table 1). Likewise, disturbed and shallow soil species (*Campanula rotundifolia*, *Danthonia decumbens*, *Galium pumilum*, *Hypericum perforatum*, *Luzula campestris* agg., *Veronica officinalis*) were commonly associated with old relevés, while ruderal species (*Impatiens parviflora*, *Rubus fruticosus* agg.) were frequently found in new relevés (Table 1).

The number of alien and invasive species also increased considerably: In old relevés, there were 1–3 alien taxa in each relevé, with just six species recorded overall. In new relevés, however, occurrence had increased to 2–4 per relevé, with 12 species recorded overall. Greatest increase was recorded in *Impatiens parviflora* which, while present in the early relevés, was only found relatively rarely (see Supporting Information Table S2).

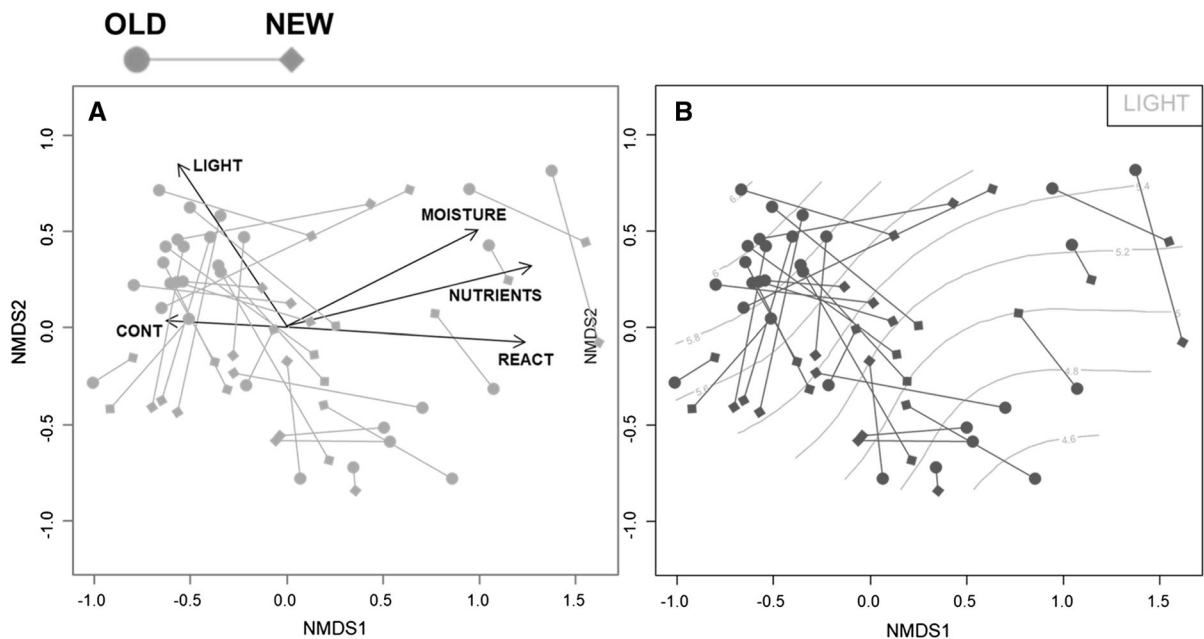


Fig. 2 Non-metric multidimensional scaling (NMDS) ordinations showing: **a** passively projected vectors of mean Ellenberg indicator values significantly ($p < 0.001$) correlated with the ordination axes, *REACT* soil reaction; *CONT* Continentality;

b passively projected mean Ellenberg indicator value for light as surface plot fitted to NMDS space. Symbols represent 29 pairs of plots (old relevés = circles; new relevés = squares)

Changes in plant diversity

In total, 202 species were recorded in the E_3 – E_1 vegetation layers in both old and new relevé datasets. Of the 202 species, 72 were found only in the old dataset (36%), 48 only in the new dataset (24%) and 82 species were common to both datasets (40%). Analysis of alpha diversity indicated a significant decrease in the number of species in the new relevés (paired t-test; $p = 0.01$). This decrease in species was also confirmed by calculations of average number of species per relevé, which showed an overall decrease of almost seven species (see Supporting Information Table S3). Vegetation heterogeneity showed no significant change over time (permutation test of multivariate dispersion homogeneity, 199 permutations, $p = 0.7$); further confirmed by the PCoA comparison of Bray–Curtis dissimilarities between relevé sets (Fig. 3).

Changes in environmental conditions

Analysis of mean EIV between old and new relevés indicated a significant change in environmental

conditions, resulting in a shift toward more shade-tolerant plant communities (Fig. 4). While the mean EIV for light showed a significant decrease ($p = 0.0419$), all other changes in environmental conditions were non-significant.

Discussion

Changes in vegetation composition

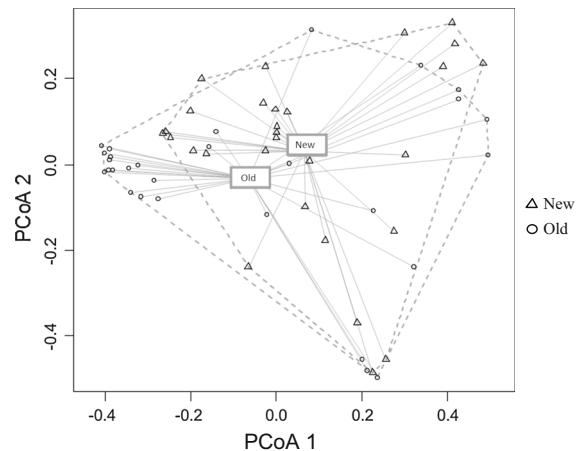
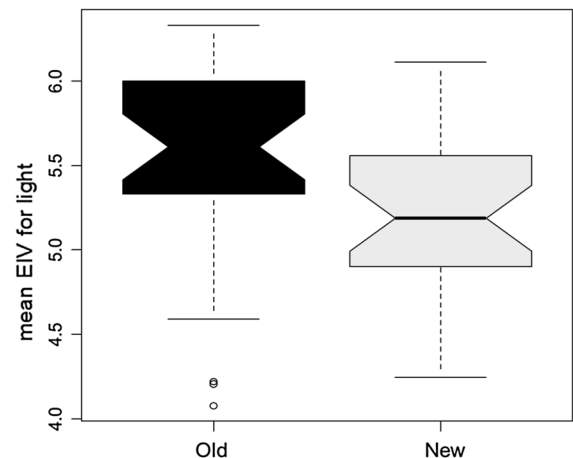
Abandonment of historical management practices, such as litter raking and coppicing, has caused an increase in tree canopy density in our study area, resulting in a shift to more shade-tolerant and nutrient-rich communities (Figs. 3, 4). A similar shift in lowland woodland character has been noted in a number of recent studies (e.g. Baeten et al. 2009; Hédal et al. 2010; Verheyen et al. 2012; Vild et al. 2013). More specifically, the increase in canopy density has prompted a decline in heliophilous species and expansion of mesophilous and shade-tolerant taxa, a parallel change also noted by Verheyen et al. (2012) and Kopecký et al. (2013). Increased understory

Table 2 Synoptic table of species fidelity (phi coefficient $\times 100$) for a set of 29 paired old and new relevés

Species	Old ($n = 29$)	New ($n = 29$)
<i>Avenella flexuosa</i>	69.8	–
<i>Hylocomium splendens</i>	51.1	–
<i>Veronica officinalis</i>	40.4	–
<i>Genista tinctoria</i>	40.0	–
<i>Calluna vulgaris</i>	40.0	–
<i>Luzula campestris</i> agg.	37.0	–
<i>Luzula luzuloides</i>	35.5	–
<i>Populus tremula</i> (E ₁)	34.0	–
<i>Ajuga reptans</i>	34.0	–
<i>Hieracium murorum</i>	32.3	–
<i>Fragaria vesca</i>	31.3	–
<i>Salix caprea</i> (E ₁)	30.7	–
<i>Hepatica nobilis</i>	30.7	–
<i>Danthonia decumbens</i>	30.7	–
<i>Galium sylvaticum</i>	30.7	–
<i>Cytisus nigricans</i>	30.7	–
<i>Hypericum perforatum</i>	30.0	–
<i>Galium pumilum</i>	30.0	–
<i>Moehringia trinervia</i>	30.0	–
<i>Campanula rotundifolia</i>	30.0	–
<i>Potentilla erecta</i>	30.0	–
<i>Veronica chamaedrys</i>	28.9	–
<i>Pinus sylvestris</i> (E ₁)	28.9	–
<i>Anemone nemorosa</i>	25.7	–
<i>Quercus petraea</i> (E ₁)	–	50.8
<i>Quercus petraea</i> (E ₃)	–	48.5
<i>Fagus sylvatica</i> (E ₂)	–	36.5
<i>Impatiens parviflora</i>	–	34.1
<i>Carpinus betulus</i> (E ₂)	–	31.1
<i>Melampyrum pratense</i>	–	30.8
<i>Quercus petraea</i> (E ₂)	–	30.7
<i>Melampyrum nemorosum</i>	–	30.7
<i>Rubus fruticosus</i> agg.	–	28.4
<i>Larix decidua</i> (E ₃)	–	27.6

Only species with the phi coefficient ≥ 0.25 are shown

shading, resulting in a reduction in heliophilous species (or reduction in overall diversity) and an increase in soil moisture (through decrease in soil surface evaporation; Liénard and Strigul 2016), is primarily due to the abandonment of coppicing and the conversion of lowland woodland to high forest (Kopecký et al. 2013). In general, the overall change

**Fig. 3** Principal coordinates analysis (PCoA) diagram representing heterogeneity between datasets. Each 'spider' and envelope connects individual sample scores with the median score (box) for the old (circle) and new (triangle) relevés**Fig. 4** Comparison of Ellenberg indicator values (EIV) for light entering old and new relevés. The plots show no overlap in the median scores (notch). The points in the old relevé dataset show outlier values, differing by more than 1.5 times of the interquartile variance (horizontal lines)

in vegetation and abandonment of coppicing in the area reflects similar trends in other Central European woodlands.

While the increase in ruderal and nitrophilous species in the area can be attributed to the end of coppice management and litter raking, a number of authors (e.g. Heinrichs and Schmidt 2017) have also suggested a connection with global environmental eutrophication, i.e. an increase in atmospheric deposition of nitrogen. Indeed, an increase in

eutrophication within lowland woodland has been noted by a number of authors, including Van Calster et al. (2007), Keith et al. (2009) and Verheyen et al. (2012). Wild game activity (e.g. mouflon [*Ovis orientalis musimon*] and wild boar [*Sus scrofa*]) causes eutrophication and ruderalisation resulting in spread of nitrophilous, ruderal, alien and invasive species (see Hédél et al. 2010; Vild et al. 2017).

Changes in plant diversity

The decrease in overall species richness (alpha diversity) was hypothesised and subsequently proven in our study area. This trend has also been observed by Kopecký et al. (2013) in other lowland woodlands and by Hédél (2004b) in mountain forests. On the other hand, other studies have either documented an increase (e.g. Van Calster et al. 2007; Šamonil and Vrška 2008; Koch et al. 2017; Reczyńska and Świerkosz 2017) or practically no change (e.g. Taverna et al. 2005; Verheyen et al. 2012) in overall species richness, suggesting there is no clear overall trend in alpha diversity shift in different forest types or locations. We suggest that species richness in lowland woodlands, at least in our study area, decreased primarily due to changes in forest management practices, which led to increased shading and a decline in heliophilous open-forest species. At the same time, the spread of deciduous trees at the expense of pines suppressed the growth of heliophilous species in the understory (Chytrý 2013).

Conversion of open forest to dense-canopied high forest tends to result in increased structural homogenisation (beta diversity) between stands, as previously demonstrated by Van Calster et al. (2007), Rogers et al. (2008), Keith et al. (2009) and Kopecký et al. (2013). Homogenisation was not demonstrated in our study area; hence the hypothesis of biodiversity loss due to homogenisation of vegetation formulated by McKinney and Lockwood (1999) cannot be confirmed in our case (see also Koch et al. 2017, who also noted unchanged beta diversity in the case of grasslands). Forest stand homogenisation accompanied by a reduction in overall species richness was also described by Hédél et al. (2010); however, they put forward a hypothesis that loss of species richness and homogenisation of forest stands can be mutually independent processes. Preservation of original heterogeneity may be supported by the activity of

ungulates such as deer or mouflon, for example, which prevent regeneration of some tree species (Hédél et al. 2010; Britton et al. 2016). In our study area, both mouflon and (especially) wild boar are relatively common and are likely to have reduced natural oak and beech (*Fagus sylvatica*) stand regeneration (Drimaj et al. 2015). We suggest that the possible slowing down of forest homogenisation by the action of wild game deserves further study.

At present, the forest vegetation in our area supports a high proportion of acidophilous species typical of nutrient-poor habitats, with the banks of streams typically supporting hygrophilous species with a high proportion of invasive nitrophytes. The majority of stands examined (not alluvial forests) demonstrate a gradual shift toward mesic conditions (i.e. increased moisture and shading, decreased understory temperatures, lower continentality, increased eutrophication), as also demonstrated by the increasing occurrence of deciduous tree species in the canopy (especially *Carpinus betulus* and *Quercus petraea*). The loss of heliophilous species, e.g. *Veronica officinalis* and *Fragaria vesca*, both previously common in the relevés (Table 1), supports the influence of shading on forest diversity, as we expected. Ongoing ruderalisation of the forest stands is also demonstrated by the expansion of ruderal species in the area (e.g. *Impatiens parviflora*, *Rubus fruticosus* agg.), supported by increased railway traffic, urbanisation associated with holiday cottages and a thriving ungulate population. The spread of alien and invasive species in the area, especially in the vicinity of settlements and roads used by heavy forestry equipment, conforms with the overall trend of environmental eutrophication increase and it is likely that these species are among the drivers that cause reduced species richness there.

Changes in environmental conditions

Our data confirmed a significant decrease in EIV for light, reflecting the increase in tree and shrub cover. Kopecký et al. (2013) also noted a significant decrease in EIV for light in lowland woodlands. We noted no other significant changes in other EIV, suggesting a relatively constant environment in terms of temperature, soil reaction, moisture and continentality. It is also possible, however, that these results reflect local changes in the stands due to the increase in tree cover

(e.g. raising humidity and reducing continentality) that may not be apparent in the total relevé dataset.

Environmental changes in lowland woodlands are usually attributed to acidification and eutrophication (e.g. Van Calster et al. 2007; Naaf and Wulf 2011; Alfonsi et al. 2016). This may be caused by an increase in atmospheric deposition of nitrogen due to industrial production or traffic (Stevens et al. 2004; Schweiger and Beierkuhnlein 2016) or changes in economic regimes that lead to an accumulation of biomass in the ecosystem (Verheyen et al. 2012).

Methodological aspects and problems related to the use of old data

Our original vegetation dataset has the advantage of being recorded by a single author over the same period. The data are also uniform in the determination of taxa, the concept of layers and species abundance, thereby reducing potential errors. Note, however, that shortcomings due to historical discrepancies in taxonomic concepts and differences in the level of knowledge regarding species determination cannot be excluded.

One of the main limitations of the historical data used in this paper is the imprecise localisation of plots by the forest territories. While acquisition of data on altitude, slope and aspect helped greatly in the determination of the original sites, frequent errors in the old dataset (e.g. altitude that cannot exist in the area) meant that only 29 of the 47 original sites recorded by V. Samek could be localised with sufficient accuracy. Similarly, Chytrý and Danihelka (1993) were only able to localise 46 of an original 80 plots in their own research.

Conclusions

Based on available methodology, historical data (forestry maps, relevé location, tree layer composition) and geoinformation technology (GIS, GPS), it was possible to obtain a relatively accurate trace of semi-permanent vegetation plots originally examined 60 years previously. To achieve maximum accuracy in location of the old sites, we also undertook a field survey into the surrounding area to reduce positional errors caused by archival map digitation.

We observed similar trends as noted in other European lowland woodlands (e.g. Becker et al. 2017), i.e. a shift toward more shade-tolerant and nutrient-rich communities, a reduction in heliophilous species abundances and alpha diversity and the expansion of alien, invasive and ruderal species. We observed no homogenisation of the forest understory, as previously recorded in similar studies, possibly due to the presence of high ungulate numbers (mouflon or wild boar) in the area preventing the regeneration of tree species.

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